

Which backgrounds should train the next BDT?

v3 signal and full Zbb/Zcc/Zss feature comparison; fixed 1,091-chunk model

FCC-ee $\Lambda_b \rightarrow \Lambda \gamma$ analysis

4 October 2026

Question and denominators

- Compare v3 offline-selected direct signal to nonmatched Zbb, Zcc and Zss before a mixed-background BDT is trained.
- Zbb: 1,091 chunks and 395,623,929 input events; Zcc: 1,200 and 499,786,495; Zss: 1,200 and 499,842,440.
- Signal: 100,002 generated direct decays in 100,000 events. Both charge conjugates included.
- Offline selection: Λ mass ± 12.5 MeV, Λ_b energy ≥ 10.5 GeV, same hemisphere. Peak: reconstructed 5.4–5.9 GeV.
- Feature figures normalize each candidate class separately. Yield figures use 6×10^{12} Z and separate processed-event denominators.

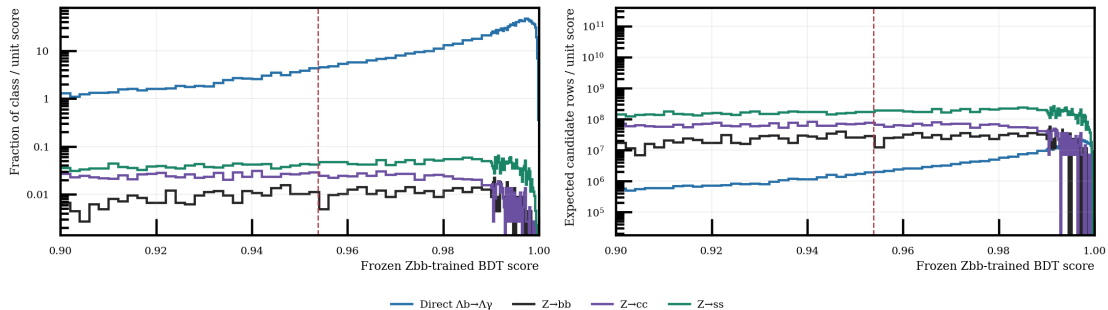
Observed candidate rows and projected peak

Class	Offline rows	BDT rows	Final peak rows	Expected final peak
Direct $\Lambda_b \rightarrow \Lambda \gamma$	54,872	37,802	23,849	194,757
Nonmatched Zbb	1,113,719	528	27	61,422
Nonmatched Zcc	1,834,608	1,720	252	363,943
Nonmatched Zss	2,161,759	4,577	514	962,511

Final means fixed score ≥ 0.9538269639 plus the proposed Armenteros, π^0 and η vetoes. The final projected inclusive peak background is 4.4% Zbb, 26.2% Zcc, 69.4% Zss under the stated Z branching scenario. These are candidate rows, not collision counts.

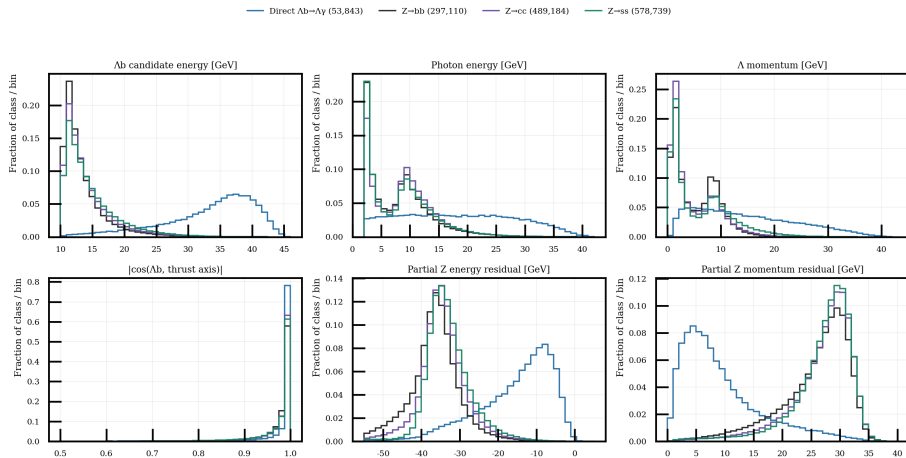
Existing Zbb-trained score tail

Offline-selected v3 candidates • score tail before veto proposals



Peak-window kinematics before BDT

v3 candidate shapes before BDT • 5.4-5.9 GeV peak

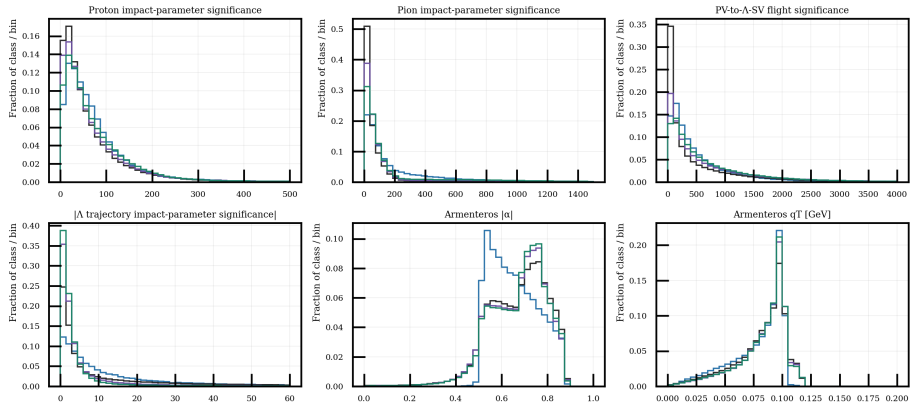


Each class normalized separately; overflow and missing values remain in denominator

Peak-window topology before BDT

v3 candidate shapes before BDT • 5.4-5.9 GeV peak

— Direct $\Lambda b \rightarrow \Lambda \gamma$ (53,843) — $Z \rightarrow b\bar{b}$ (297,110) — $Z \rightarrow c\bar{c}$ (489,184) — $Z \rightarrow s\bar{s}$ (578,739)

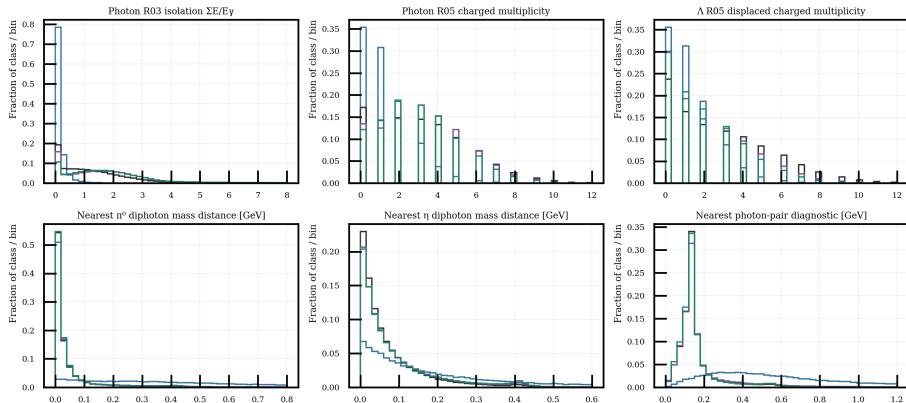


Each class normalized separately; overflow and missing values remain in denominator

Peak-window isolation and photon pairing

v3 candidate shapes before BDT • 5.4–5.9 GeV peak

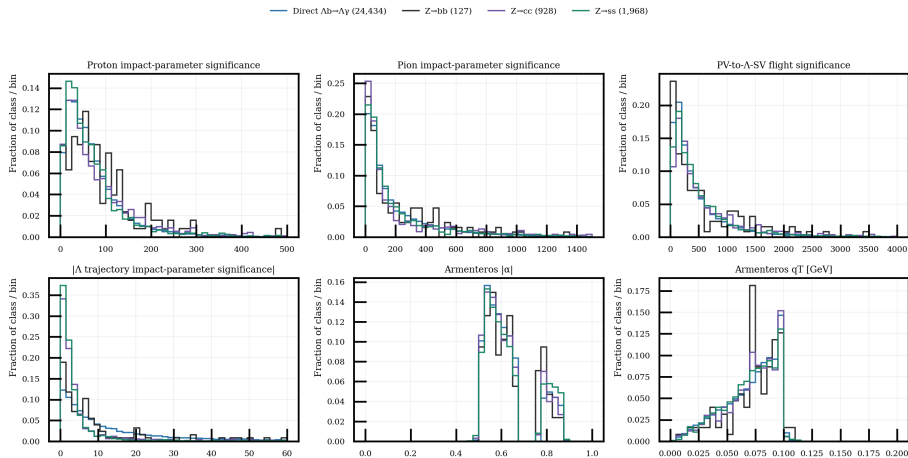
— Direct $\Lambda_b \rightarrow \Lambda \gamma$ (53,843) — $Z \rightarrow b\bar{b}$ (297,110) — $Z \rightarrow c\bar{c}$ (489,184) — $Z \rightarrow s\bar{s}$ (578,739)



Each class normalized separately; overflow and missing values remain in denominator

Topology in the current surviving tail

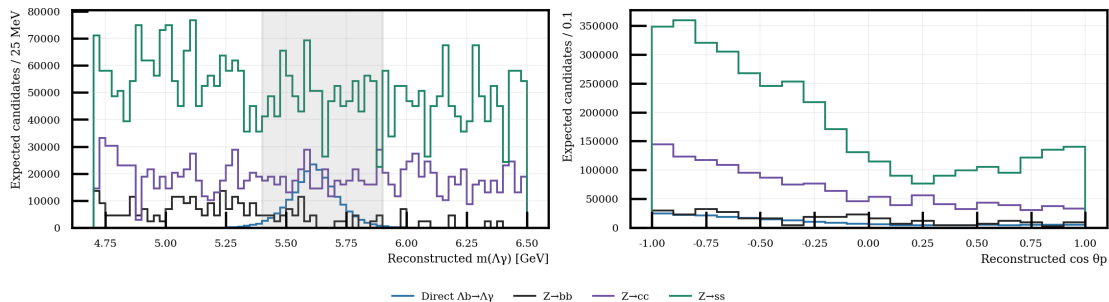
v3 candidate shapes after BDT and veto proposals • full 4.7–6.5 GeV fit interval



Each class normalized separately; overflow and missing values remain in denominator

Final expected mass and angle on a linear scale

v3 after BDT and veto proposals • linear physical projection, 6×10^{12} Z



Proposed controlled model comparison

- Compare the same 20 features with Zbb-only training and with a physically motivated Zbb/Zcc/Zss background mixture.
- Add reconstructed $|d_0(\Lambda)/\sigma|$ as a separate mixture-model ablation. It is absent from the current 20 features; its peak marginal TV from signal is 0.415 for Zss versus 0.189 for Zbb.
- Keep Armenteros α, q_T as a separate test. Existing energy, recoil and isolation features already carry strong signal/background separation.
- Freeze train/validation/test by event and production chunk. Optimize $S/\sqrt{S+B}$ in the peak on validation, then check per-flavour tails, mass/angle and PHSP acceptance on held-out data.
- These plots are descriptive and include existing Zbb training chunks. No mixed model or new feature is adopted here.